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RESEARCH ARTICLE

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Green industrial policies: opportunities and obstacles from the global trade and investment regime

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ABSTRACT

Many countries are returning to industrial policy tools to meet various growth and development goals, such as participating in newly emerging global value chains, ensuring supply chain resilience, and adopting renewable energy sources. At the same time, the existing stock of trade and investment agreements may act as an obstacle to industrial policymaking. The rules in these agreements embody a more "hands-off" approach to economic governance that prioritizes nondiscrimination and the removal of as many restrictions on cross-border economic activity as possible. This paper analyzes the extent to which countries face actual or potential legal constraints in their pursuit of green industrial policy under the international trade and investment regime. To answer that question, a green industrial policy landscape is created from existing policy data. The paper then examines the extent to which those policies are actually or potentially threatened by current trade and investment rules governing industrial policy measures.

Consistent with other related literature, the research finds that advanced economies are the most active policymakers. However, the policy mix varies significantly between advanced economies and emerging market and developing economies, with the latter tending to rely on measures that are more strictly constrained by rules.

Despite these constraints, the importance of globally inclusive participation in climate action is imperative, and newly emerging global value chains for low-carbon technology products may provide opportunities for lower-income countries to promote growth through green industrial development. The tension, therefore, must be resolved to ensure that trade and investment rules do not pose an obstacle to this new wave of industrialization that contributes to the global goal of slowing climate change for the benefit of current and future generations.

Key policy insights

- The mix of policy tools used by advanced economies and emerging market and developing economies differs, but even advanced economies are increasingly using policies prohibited by global trade rules.
- Most countries rely heavily on permitted policies, but this does not fully account for the ways nondiscrimination and nonarbitrariness standards constrain public policy.
- Policies enabling the development of green supply chains must be treated deferentially under trade and investment rules.

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Introduction

Many countries are looking to the future with the hope of successfully transitioning to become low-carbon economies. Long-term structural transformation toward more climate-conscious consumption and production is as much a priority of climate policymaking as mitigation and adaptation, emergency response, and recovery from loss and damage (Pelling et al., 2015). Countries cannot slow down the effects of climate change without immediate and rapid mitigation of carbon emissions, while also helping populations adapt and respond to the present-day climate realities. While these immediate and reactive goals may suffice in the short term, true resilience depends on countries deploying policies that restructure entire economies toward a low-carbon future. Therefore, a successful climate policy strategy would involve both immediate mitigation policies and industrial policies that would affect long-term structural change (Climate Action Network, 2020; NDC Partnership, n.d.; Stiell, 2024).

Industrial policy has long been identified as a subset of national policies that aims to shape or transform the structure or composition of economic activity. Although the benefits and drawbacks of such policies have been hotly contested, a diverse set of countries are turning to industrial policy tools as a way to respond to geopolitical pressures and shore up domestic industrial sectors as well as address health and environmental crises. Notably, countries that have previously been critical of industrial policies have shifted their stance. The US' Inflation Reduction Act and the CHIPS and Science Act, as well as the European Union's (EU) Green Deal, provide some examples (DeFazio, 2020; Ryan, 2022). This renewed interest in industrial policy is not limited to advanced countries, as evidenced by national industrialization plans (IEA, 2024). These tend to include policies, referred to as 'green' industrial policies, which support climate-related sectors like renewable energy and electric mobility.

Nevertheless, many modern trade and investment agreements, including the World Trade Organization (WTO) agreements, were negotiated and drafted at a time when industrial policies had fallen out of favour in the advanced economies (AEs) that largely dictated the rules of such treaties (Shadlen, 2005). These treaties embrace a more 'hands-off' approach to economic governance that prioritizes nondiscrimination and the removal of as many restrictions on cross-border economic activity as possible (Thrasher & Gallagher, 2010). This sentiment exists in tension with industrial policy as it calls for a 'hands-on' approach to economic governance where the state directs resources to industries that it deems are of strategic importance.

In light of this tension, this paper examines the extent to which countries face actual or potential legal constraints in their pursuit of green industrial policy (GIP) under the international trade and investment regime. To answer that question, the paper first establishes the extent to which countries are deploying GIPs to create a current policy landscape. Second, it examines the extent to which those policies are actually or potentially threatened by the current trade and investment rules by creating a comprehensive mapping between these industrial policy actions and trade and investment rules.

The following sections of the paper review the literature on how trade and investment treaties constrain domestic policymaking and on new trends in industrial policy. The paper then locates this research within a system of national and international obstacles to GIP through a conceptual framework. Subsequent sections explain the data and methodology for the analysis, the findings of that analysis, and the discussion of related issues for further consideration. The paper concludes by laying out three possible ways forward for policymakers navigating the tension between GIP practices and global trade and investment rules.

Review of the literature

The literature exploring the tension between trade and investment treaties and domestic policymaking supports the conclusion that countries' commitments as members of the WTO, bilateral and regional free trade agreements (FTAs), and international investment agreements (IIAs) have substantially curtailed countries' policymaking ability (Thrasher, 2021). This has impacted policies supporting public health (Baker & Thrasher, 2023; Gleeson et al., 2019; Tenni et al., 2022), fiscal and financial management (Dutt et al., 2020; Thrasher et al., 2021), environmental protection (Tienhaara, 2018; Yamaguchi, 2020), and general industrial development (Thrasher & Gallagher, 2010).

As climate policy has become more common, the same question arises: to what extent are countries free to adopt climate policy under their current trade and investment commitments? Research has demonstrated the large financial risk that investor-state dispute settlement (ISDS) poses to countries attempting to phase out

fossil fuel extraction and production (Tienhaara et al., 2022b). Other research has analyzed the legal challenges that Canada, India, and the US have faced for deploying local content requirements to promote the development of renewable energy sectors (Thrasher, 2019). There have also been studies identifying inconsistency between specific climate policy proposals and trade and investment rules, and proposing how to shape the policies to ensure compliance (Bacchus, 2021; Nam, 2023).

The second body of literature attempts to analyze the upward trend in industrial policy drawn from trade policy data developed by the Global Trade Alert (GTA) (Evenett et al., 2024; Juhász et al., 2023; Lewis, 2021). The GTA team created the New Industrial Policies Observatory, which identifies a subset of its trade policies as 'industrial policies', according to the stated motive and the products or sectors affected, among other features. Juhász et al. (2023) rely on natural language processing algorithms to identify the intent of the policy to isolate industrial policies from the broader GTA database. An older study undertaken by Lewis (2021) identified industrial policies by using keyword searches in the policy descriptions.

This paper unifies these separate lines of investigation by creating a mapping between the industrial policy actions identified in the GTA database and trade and investment rules that may constrain policy space. By doing so, it responds to Evenett et al. (2024, p. 25), who note the importance of exploring the extent to which current trade rules are 'up to the challenge of … allowing sufficient [policy] flexibilities when needed'. To directly answer this question, this paper narrows the focus on GIP and investigates the extent to which the landscape of policies deployed over eight years (2016–2023) is permitted by the existing trade and investment treaties.

Consistent with other related literature, this paper confirms that AEs are the most active policymakers (Evenett et al., 2024; Juhász et al., 2023; Thrasher et al., 2023). It also finds that the policy mix differs significantly between AEs and emerging market and developing economies (EMDEs), with AEs relying primarily on subsidy measures and EMDEs relying more heavily on trade-restrictive measures and local content requirements. Moreover, the policies preferred by AEs enjoy more deferential treatment under trade and investment rules, whereas EMDEs tend to rely on measures that are more tightly constrained under those rules. The discussion further highlights the outsized impacts of general standards of nondiscrimination and nonarbitrariness in trade jurisprudence and notes significant differences in impact between trade and investment jurisprudence.

Conceptual framework

Currently, the extent of use and type of industrial policies are as diverse as the countries that deploy them (Fasteau & Fletcher, 2024; Juhász et al., 2023; Thrasher et al., 2024). Furthermore, as illustrated in Figure 1, GIP agendas are now being viewed as serving multiple development and growth-related goals of countries, such as building supply chain resilience, export-driven growth, creating new and more skill-intensive jobs, climate change mitigation, and adaptation. Yet for many years, industrial policies were not accepted as a viable policy tool and were criticized for promoting rent seeking, state capture, and resource misallocation by the neoliberal economic order (Rodrik, 2009).

Beyond ideological concerns, the implementation of industrial policy, especially GIP, may face several practical barriers. These could be at the national level and may include limited fiscal space, competing development priorities, government capacity-related issues, and a lack of technological know-how to kick-start industrialization in sophisticated low-carbon technologies. Additionally, there could also be barriers at the multilateral level, including geopolitical tensions, economic fragmentation, unilateral policymaking by wealthier countries, and international trade and investment rules. This paper examines this last channel.

As noted above, many studies anecdotally highlight how trade and investment rules may discourage specific kinds of policies or rein in the negative economic effects of unilateral policies (Bacchus, 2021; Nam, 2023; Thrasher, 2019; Tienhaara et al., 2022b). In contrast to these case-specific analyses, this paper is the first to assess the extent to which the whole landscape of GIP may be in tension with countries' trade and investment commitments. It does so by mapping trade and investment rules onto past industrial policy action at the country level. By doing this, it establishes whether these trade and investment rules pose a credible barrier to green industrial policymaking by considering the following scenarios:

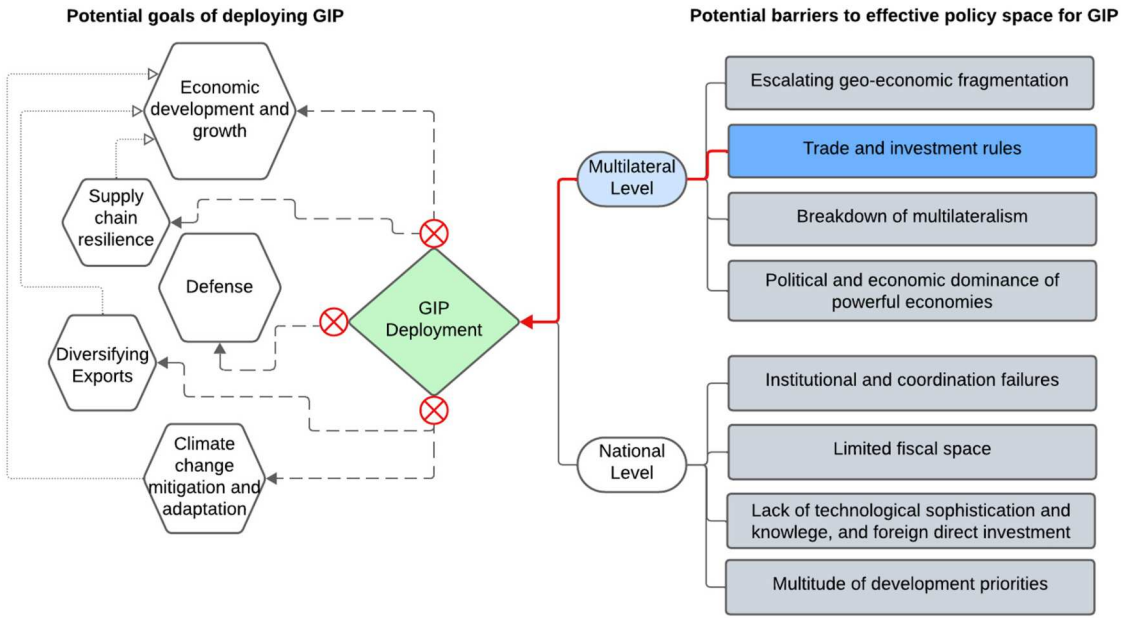


Figure 1. Trade and investment rules as barriers to GIP policy space. Source: Authors' illustration.

1. If all industrial policies are within the rules, either countries are adapting their policymaking to the rules or the rules do not constrain current policymaking practices.
2. If countries are implementing policies that are against the rules but are not being contested, then the rules are ineffective and are possibly not a barrier to industrial policymaking.
3. If there is heterogeneity in the kinds of policies that different groups of countries are implementing, with one group 'staying within the rules' and the other not, it might indicate that these rules are constraining one set of countries more than another.

If the evidence points to case two and/or three, policymakers need to ask whether the current set of rules is 'fit for purpose'. However, if the evidence points to the first case, it may be that in the particular case of GIP, trade and investment rules are not a barrier to policy space.

Data and methodology

To create a global landscape of GIP, this paper draws on trade policy data from the GTA database. As described below, the GTA data allow researchers to compute descriptive statistics on the types of policies, products, and sectors affected, as well as the countries implementing them. These data are then combined with legal standards drawn from trade and investment treaties to assess whether the identified policies fall within the constraints of current global rules.

Data

The sample of countries included in this study is the same as the number analyzed by the GTA and includes 40 AEs, 154 EMDEs, and China.[1] The period of analysis is from 2016 to 2023. Although the starting year of the analysis is somewhat arbitrary, this window reflects the rise of GIP deployment. Following the literature,

[1]Note that the final sample in the analysis is further limited to the countries that enacted policies related to one or more of the selected sectors.

this paper only analyzes the red-coded policies (red alerts) of the GTA. A red alert is an action a country takes that, according to the GTA, almost certainly discriminates against foreign commercial interests (Evenett, 2019).[2] These conditions fulfil the definition of industrial policy actions described in the introduction as any government action that aims to deliberately shape or transform the structure or composition of economic activity.

To select the industrial policies, of the five sectors, the paper uses the Harmonized System (HS) product codes from the International Monetary Fund (IMF) list of low-carbon technology goods (Arslanalp et al., 2023) and the list of Minerals for Climate Action (Hund et al., 2023). GTA policies are listed on a product-by-product basis and are also classified according to the HS. This allows the selection of a subset of red alerts aimed at those industries. The sample is further narrowed based on categorical exclusions, such as policies with economy-wide effects (e.g. Generalized System of Preferences beneficiary changes), policies aimed at security or geopolitical concerns (e.g. additions to the US' entity list), policies related to COVID-19 (by keyword search), and policies impacting a large number (more than 200 'affected products') of sectors that are not limited to 'green' products. The resulting panel dataset of industrial policies at the industry level by country and by year is then merged with country characteristics such as income and country group based on the World Bank and World Economic Outlook classification of the IMF.

The policy categories include export and import policy instruments, foreign investment policy, localization policy, public procurement policy, subsidies and state aid, and trade defense, based on the GTA's existing policy categories, for ease of exposition (see Appendix Table A1).[3]

Sectoral selection

The paper focuses on five sectors – critical minerals, batteries, solar power, wind power, and electric vehicles (EVs) – which were selected due to their relative importance in international trade flows, the frequency with which they are referenced in the literature, and the potential that countries have to move up the value chain toward a deeper, long-term structural transformation. In terms of trade volumes, products related to solar power, wind power, and EVs are among the most widely traded low-carbon technology products (Bandara et al., 2025; see Appendix Figure A5) and are frequently the targeted industry for climate policies. The industry-oriented literature has likewise zeroed in on these strategic sectors, as indicated in the most recent International Energy Agency report, which principally covers these three industries (IEA, 2023). Moreover, both the battery and critical minerals industries are inextricable from solar, wind, and EVs, given their role in the value chains for each of them.

Mapping trade and investment rules and cases

For the purposes of this paper, the global rules governing international trade and investment include the WTO, the more than 350 FTAs notified to the WTO and the almost 2000 standalone IIAs in force worldwide. These rules aim to minimize obstacles to cross-border economic activity and eliminate discrimination between and among trading partners. In general, these two principles act as 'screens' to filter out any

[2]For an observation to be classified as a red alert in this way, the GTA team passes the observation through a series of tests, which include looking for language that implies a particular intervention is designed for trade discrimination or designated as fulfilling national competitive interests.

[3]It is important to note that the GTA does not track changes to a country's intellectual property laws, which have been identified as an important category of industrial policies to encourage both innovation and dissemination of essential climate technologies (Correa, 2016; Fink & Raffo, 2019). Policies related to intellectual property rights may include changes to intellectual property protection (either loosening or tightening up that protection), issuing compulsory licenses for the generic production of inventions and written works, and introducing policies for data protection. Some experts have identified changes to intellectual property rights policies as key to facilitating a just energy transition (Deane & Hurst, 2024; Maskus & Okediji, 2010). It is also true that the EV sector is experiencing a surge in patent applications alongside the surge in EV-specific industrial policies like subsidies (Energy Policy Institute, 2024). However, more research must be done to determine whether countries are seeking to remove intellectual property rights policies obstacles to facilitate the diffusion of essential climate technologies, a move that has been advocated in the 'Access to Medicines' movement over the past 40 years (e.g. Hoen, 2002).

policies that might contravene the goals of the system – freer and fairer trade among all nations (Thrasher et al., 2024). In tandem, the rules seek to minimize arbitrariness or a lack of due process in the process of governing economic activity. Almost all global trade and investment rules fall under one or more of these three principles.[4]

To map current policies onto specific trade and investment rules, each instrument type receives a score according to whether they are prohibited, regulated, or permitted. A policy instrument with a '2' score is *prohibited* under trade and investment rules. These provisions are marked by clear rules indicating that 'no member shall apply' such a measure or that 'no [such measure] shall be instituted or maintained'. A policy instrument with a '1' is allowed but *regulated* in specific ways – either through scheduling (e.g. tariffs and tariff quotas) or through notification/transparency requirements (e.g. licensing requirements) or is challengeable in the case of negative economic impacts on other member states (e.g. actionable subsidies). A policy instrument with a '0' is *permitted* and constrained only by the most basic nondiscrimination and nonarbitrariness standards, though exceptions may apply (see below). From that score, the proportion of each country group's policies that fall into the 'prohibited category' is calculated, as well as average scores for the landscape of policies for each country group over time. This process assesses (1) which group is more likely to introduce policies that violate trade and investment rules and (2) whether that policy practice has been consistent or changed over time.

Findings

The landscape of GIP

The number of industrial policies ('interventions' or 'acts') aimed at solar, wind, EVs, batteries, and critical minerals industries (henceforth GIP) has steadily increased since 2020 (see Appendix, Figure A1). In 2023, the share of GIP was around 20 percent of all industrial policies enacted by all countries in the study. Furthermore, growth in GIP is observed in every sector except EVs, as countries appear to be shifting toward battery and transition materials development (see Appendix, Figure A2).

Like many industries that saw an increase in industrial policy action in 2020, the battery industry alone had more than 200 acts that year (in 2019, this number for batteries was around 40 acts). Even though batteries have not received comparable action since, this sector has remained a central focus among most countries. This focus makes sense given the need for batteries in EVs as well as in solar and wind generation to solve intermittency issues. Analyzing the battery industry by income groups shows that, as with most other industries, high-income countries are the most active policymakers (see Appendix, Figure A3). What stands out, as documented in this dataset, is the absence of almost any GIP measures by low-income countries in any of these industries, though there are some exceptions to this in the critical minerals industry. For example, the Democratic Republic of the Congo enacted an export ban to support cobalt and copper extraction in 2021. Lower-middle income countries are also actively engaging in the EV industry, with India accounting for about 10 acts to boost its domestic EV sector as part of a government effort to promote electric mobility (Government of India, n.d.). Despite these positive developments, policy action is driven predominantly by advanced countries and China (which accounts for most actions by the upper-middle income group).

Although the category of industrial policy that is most used across the study sample is 'Subsidies and state aid' (see Appendix, Figure A4), some consequential patterns emerge by breaking down policy categories by country group. Figure 2 shows the breakdown of the aggregated policies that countries enacted from 2021 to 2023. In total, AEs, China, and EMDEs enacted 740, 252, and 230 acts, respectively. Most policies adopted by AEs are of the 'Subsidies and state aid' category (more than three-fourths), and China relies almost entirely on this category as well. By contrast, nearly half of the policy toolkit of EMDEs (excluding China) comprises 'export and import policies' and around 10 percent on local content policies. As discussed in more detail below, this policy mix is far more likely to be expressly prohibited under global trade rules, which could expose EMDEs to a higher risk of disputes challenging their policies.

[4]Beyond these general principles, there are also specific rules in both trade and investment agreements that prohibit certain measures altogether and that regulate others by limiting their use.

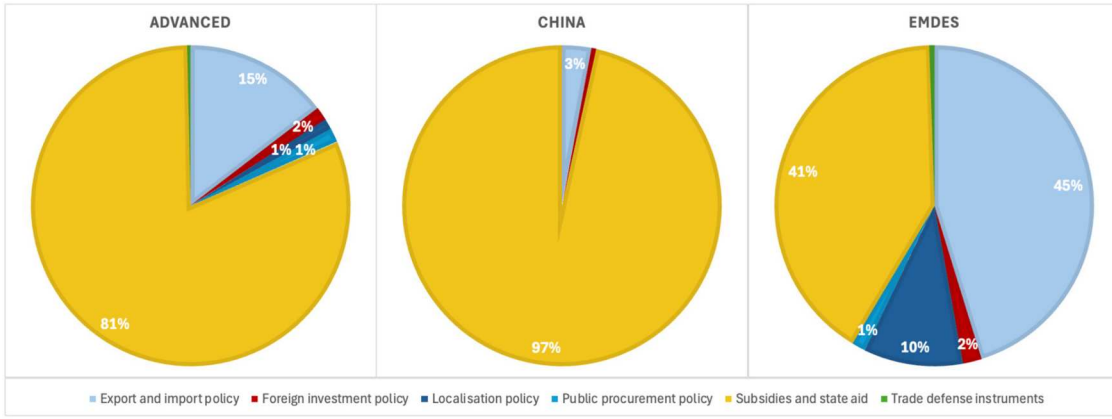


Figure 2. Industrial policy by country group, 2021–2023. Source: Authors' calculation.

Mapping the legal risk of the GIP landscape

Considering the heterogeneity of the policy toolkit of different country groups, this section explores the extent to which global trade and investment rules constrain the use of those policies. The analysis shows that the intervention categories with the highest number of 'prohibited' policies are export and import policies and localization policies (Table 1). Both types are proportionally used more by EMDEs than by AEs (Figure 2). Thus, although AEs are much more active policymakers as a group, the rules tend to treat their policy mix more deferentially.

This finding is confirmed by Table 2, which identifies the proportion of policies that are prohibited for each country group in 2023. A 'prohibited' policy (any intervention that has a score of 2) is labelled 'Yes' in the table, and 'No' includes policies with scores of both 0 and 1. The calculations show that nearly 15 percent of all EMDE policies violated rules in 2023 compared with a mere 3 percent of AE policies. Although the value for China is unusually high at around 11 percent, Figure 3 shows that China generally tends to deploy more rule-compliant policies.

Figure 3 applies the scoring system in Table 1, where each GIP receives a score of 0 if permitted, 1 if regulated, and 2 if prohibited. The total score for each country group is then divided by the number of GIPs to obtain an average score. When interpreting this figure, a value closer to 2 would indicate higher utilization of prohibited policies, while values closer to 0 indicate compliance. The figure shows that since 2017, EMDEs have consistently had a higher score than both AEs and China (except in 2022), further reinforcing the argument that the policy mix of EMDEs is more likely to be challenged compared to that of AEs. Notably, although AEs' average score is below 1 for the entirety of the period of analysis, it has been trending upward since 2021, suggesting an increased reliance on policies that are more tightly regulated (or even prohibited) by trade and investment rules.

Discussion

Once more, the data uncovered significant heterogeneity in the policy toolkits between AEs and EMDEs such that the policies preferred by EMDEs are those that have been challenged more frequently in WTO jurisprudence (see Appendix Table A2). It is important to note that legal challenges do not appear instantaneously upon the introduction of the offending policy, and it is therefore infeasible to show a direct empirical relationship between prohibited policies and related legal challenges.[5] It is also notable that China has emerged as a major player in new climate-related cases (see Appendix Table A2).

[5]It is beyond the scope of this paper to assess all the potential other kinds of legal, diplomatic, or political challenges that countries may have faced as a result of their policymaking, whether through domestic court cases, state-to-state consultations under FTAs, or more informal mechanisms.

Table 1. Intervention types and rules compliance.

Intervention category	Permitted – 0	Regulated – 1	Prohibited – 2
Export and import policy	1. Export tax 2. Export-related nontariff measure, nes 3. Other export incentive 4. Import monitoring 5. Internal taxation of imports 6. Import-related nontariff measure, nes 7. Foreign customer rule 8. Trade finance 9. Trade payment measure	**GATT Art. II:** 1. Import tariff 2. Import tariff quota **ILA.** Import licensing req't **SCM Art. 5** Import incentive	**GATT Art. XI**: 1. Export ban 2. Export licensing req't 3. Export quota 4. Export tariff quota 5. Import ban 6. Import quota **SCM Art. 3**: 1. Export subsidy 2. Tax-based export incentive **TRIMs Illustrative Annex:** Trade balancing measure
Foreign investment policy	FDI: Treatment and operations	**GATS Schedules**: FDI: Entry and ownership **SCM Art. 5**: FDI: Financial incentive	
Localization policy	1. Local labour incentive 2. Local operations requirement*		**TRIMs Illustrative Annex/SCM Art. 3**: 1. Local content incentive 2. Local content requirement 3. Local supply req't for exports 4. Local value-added incentive 5. Local value-added req't 6. Localization, nes
Public procurement policy		**GPA**: 1. PP access 2. PP localization 3. PP preference margin 4. PP, nes	
Subsidies and state aid		**SCM** 1. Capital injection and equity stakes (incl. bailouts) 2. Financial assistance in foreign market 3. Financial grant 4. In-kind grant 5. Interest payment subsidy 6. Production subsidy 7. Price stabilization 8. State loan 9. Loan guarantee 10. Tax or social insurance relief 11. State aid, unspecified 12. State aid, nes	
Trade defense instruments	1. Anti-circumvention	**GATT Art. VI/ADA** Anti-dumping **GATT Art. VI/SCM** Anti-subsidy/countervailing duty **GATT Art. XIX, SG** Safeguard **Agreement on Agriculture** Agriculture Special Safeguard	

*Explicitly permitted in United States-Mexico-Canada Agreement USMCA (2020) Art. 14.10.3.
Note: GATT 1947: General Agreement on Tariffs and Trade (1947); GATS: General Agreement on Trade in Services (1994); SCM: Agreement on Subsidies and Countervailing Measures (1994); ILA: Agreement on Import Licensing Procedures (1994); SA: Agreement on Safeguards (1994); TRIMs: Agreement on Trade-Related Investment Measures (1994); GPA: Agreement on Government Procurement (2012).
Source: Authors' illustration.

The question of why this heterogeneity exists has not been fully explored. Some research has rightfully pointed out that subsidies often require fiscal space not available for many EMDEs (Juhász et al., 2023). Moreover, since the WTO's Appellate Body has been wholly or partially disabled during the period of analysis, it is reasonable to speculate that the lack of effective enforcement may encourage EMDEs to deploy more

Table 2. Policy interventions and trade and investment rules mapping, by country category and sector.

	Clear violation (score = 2)	Solar	Wind	EV	Battery	CRTM	% Violations of Total GIP
Advanced	No	61	37	23	125	53	2.6%
	Yes	**5**	**1**	**1**	**1**	**0**	
China	No	12	1	2	6	4	10.71%
	Yes	**1**	**0**	**0**	**1**	**1**	
EMDE	No	12	11	18	18	5	14.67%
	Yes	**1**	**3**	**3**	**2**	**2**	

Source: Authors' calculation.

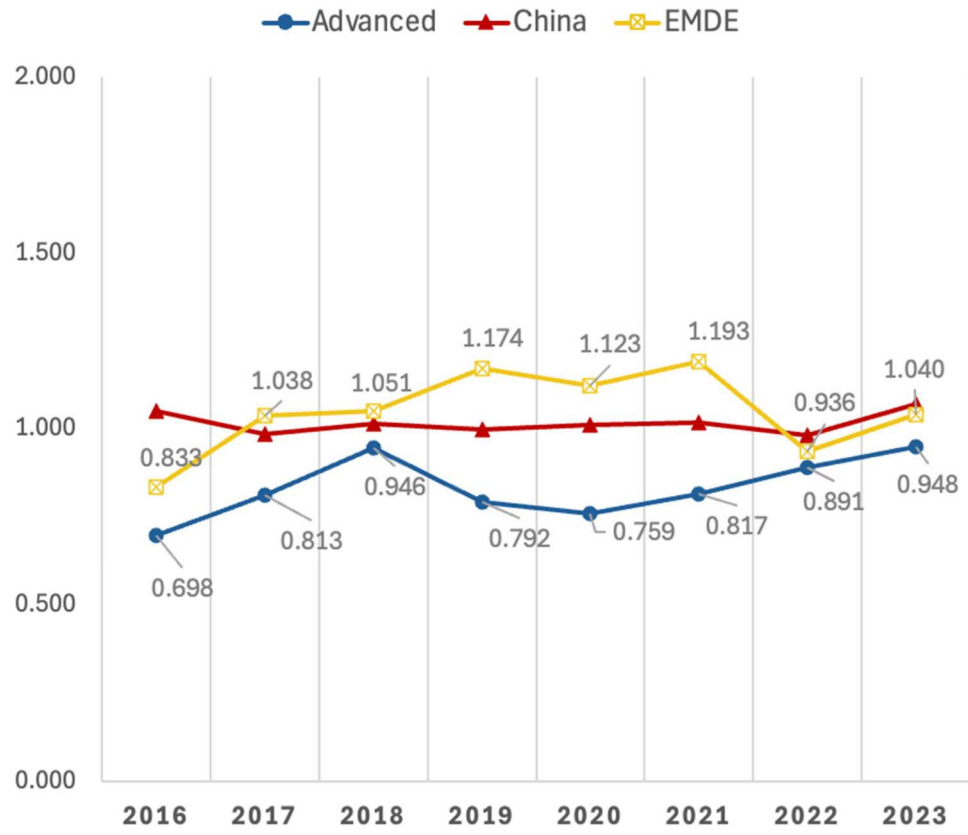


Figure 3. Average policy scores by country group. Source: Authors' calculation.

'prohibited' GIP. It is clear from the presence of prohibited policies that the rules and associated enforcement measures do not demand perfect compliance. At the same time, there is a relatively high percentage of policies, even in EMDEs, that are 'permitted' or 'regulated' under the rules (85 percent in 2023 for EMDEs), suggesting that the rules allow a substantial diversity in policymaking and many countries can and do operate within them.

However, the prohibited-regulated-permitted trichotomy does not tell the whole story. In particular, by not counting the impacts of the nondiscrimination or nonarbitrariness standards, this approach fails to account for the outsized role these standards have played in trade jurisprudence. For the specific context of GIP, where developing domestic industries is an essential component, nondiscrimination standards may be an even bigger obstacle. The scoring system also does not account for differences between trade and investment jurisprudence in terms of (1) hurdles to bringing a case, (2) legal certainty about how the standards will be applied, and (3) the costs of losing a case.

Nondiscrimination and nonarbitrariness in WTO jurisprudence: an outsized impact

A brief exploration of WTO jurisprudence demonstrates both that (1) public policies that do not violate the express prohibitions in the treaties may still violate nondiscrimination standards, and (2) public policies that may appear to be justified under the general exceptions (Article XX) almost never get past the strict construction of the nondiscrimination and nonarbitrariness clauses.

Interpreting and enforcing standards of nondiscrimination has been integral in WTO cases involving public interest policies, where health or environmental policies were adopted, resulting in effective discrimination against competitive imported products. Even the cases invoking express prohibitions in the rules (e.g. GATT Article XI and the TRIMs Agreement; see Table 1) often involve general nondiscrimination standards. In the *EU-Palm Oil* cases, for example, the measures were not found to violate Article XI, but they did violate the most-favored nation (nondiscrimination) standard (Art. I) in effect. WTO panels have consistently invoked the presence of discriminatory effects (or de facto discrimination) as a reason to find a violation of the rules even where the measure was neutral with respect to nationality (Ehring, 2001, 2014). Combined, the national treatment and most-favored nation treatment standards make up 30 percent of the treaty provisions invoked in WTO dispute settlement (Hoekman et al., 2021).

The prohibition of arbitrary state behaviour in the WTO rules can be found (among other places) in the preambular language of the General Agreement on Tariffs and Trade (GATT) Article XX. This paragraph states that all justifiable public purposes for violating the GATT are 'subject to the requirement that such measures are not applied in a manner which would constitute a means of arbitrary or unjustifiable discrimination … or a disguised restriction on international trade'. In other words, countries must not only have good public purposes for violating a rule but also must *not apply* the measure in a way deemed arbitrary, unjustifiable, or lacking good faith (a 'disguised restriction').

In practice, only two cases have succeeded on Article XX grounds: *EC–Asbestos* and *US–Shrimp* (Art. 21.5 compliance proceeding) (Mitchell et al., 2017; Pelton, 2024; Voon, 2022).[6] Other public policies that measured up to this standard have all been found wanting due to their discriminatory implementation.[7]

Fundamental differences between trade and investment disputes

There are also relevant distinctions between the regimes of trade and investment jurisprudence. First, trade jurisprudence is state-to-state only, meaning that states ultimately decide whether to bring a case (and may have various reasons for not doing so). By contrast, investment commitments often allow foreign investors to bring a claim directly against their host state when new policies or regulations affect the value of their investment. These investors are not beholden to the public but instead only to their own shareholders, which significantly lowers the hurdle for initiating claims.

Second, just as general standards of nondiscrimination and nonarbitrariness are much more often invoked in trade disputes than specific rules, this is even more true in ISDS. Broad rules like those governing 'indirect' expropriation and the 'fair and equitable treatment' standard are invoked with much higher frequency than specific rules. States also lose at higher rates in those cases (see Appendix Table A3). However, this risk is not captured by the scoring above because most policies in the GTA's 'foreign investment measures' category are not prohibited under IIAs.[8]

[6] *EC–Asbestos* (2001) involved a public health measure to ban the use of asbestos in building materials and the import of asbesto-containing products. The *US–Shrimp* (2001) case involved a ban on imports of shrimp based on how traditional fishing methods posed a danger to dolphins. In the latter case, the US lost at first and had to reconfigure the measure so that it did not result in de facto discrimination.

[7] See, for example, *Brazil–Tyres* (2007) (because the ban did not apply to the MERCOSUR countries) and *EC–Seal Products* (2014) (because the restrictions did not apply to certain Indigenous groups in Greenland but did apply to Indigenous groups in Canada and Norway). While in those cases, the measures were theoretically justifiable under one of the public policy purposes laid out in the sub-paragraphs of Article XX, they did not make it past the strict construction of the preamble as interpreted by the WTO's Appellate Body. Recent climate-related jurisprudence, as in *EU–Palm Oil*, seems to follow this trend. While the EU's measures were deemed justifiable under Article XX, they did not succeed on the basis of the nondiscrimination and nonarbitrariness standards of the preamble.

[8] This may seem particularly surprising given that recent studies have shown the large financial risk faced by countries attempting to phase out fossil fuel exploration, extraction, and production (Tienhaara, 2018; Tienhaara et al., 2022a, 2022b). While the potential for these claims continues to grow, so far they have been rare.

Furthermore, ISDS cases frequently challenge the *process* involved in government action. Governments may face internal logistical challenges, domestic political opposition, or international pressure that require repeated decision-making changes. In *Rockhopper Mediterranean Ltd v. Italian Republic* (2024), the case ultimately turned on slow, tortuous, and delayed government decision-making in response to public pressure. Likewise, Colombia faced multiple legal challenges to its iterative process of delineating the protected *páramo* areas and subsequently banning mining there (*Montauk Metals v. Colombia* (2024), *Red Eagle v. Colombia* (ICSID, 2024), *Eco Oro v. Colombia* (2021)).[9] In short, this demonstrates that even generally permitted policies are not shielded from investor challenges.

Third, the costs of losing a case before the WTO are far lower than the costs of losing an investor-state dispute. Under the WTO Dispute Settlement Understanding (DSU), losing respondents must bring their laws into compliance with the rules or face equivalent retaliatory measures (DSU, 1994, Art. 22). When all is working properly, there is an appeal process by which the losing party can contest the outcome. In ISDS, however, the only remedy available is a monetary award, whose magnitudes have been criticized for their excessiveness and crippling impact on respondent states, and there is no opportunity for appeal (Bonnitcha & Brewin, 2020; Moehlecke & Wellhausen, 2022; Paparinskis, 2020; Tienhaara et al., 2022b).

Nondiscrimination, environmental protection, and equity

The above findings and discussion illuminate the extent to which current GIP practice is in tension with existing trade and investment rules, including general principles of nondiscrimination and nonarbitrariness. They also highlight the potential legal and financial risks associated with that tension. However, they do not preclude an important and sometimes positive role that nondiscrimination standards play in global trade and investment governance. Indeed, in some of the cases described above, the goal of applying a nondiscrimination standard was to encourage better protection of the environment overall. At the same time, environmentally motivated discriminatory policies introduced by AEs can expose lower-income countries to negative economic externalities (He et al., 2022; Titelman et al., 2023; UNCTAD, 2021) or result in inappropriate external policy imposition. This concern arose in many early environmental cases (e.g. *US–Tuna*; *US–Shrimp*) and can still be seen in the *EU–Palm Oil* cases.[10] These debates around whether nondiscrimination standards, as applied, should prioritize environmental protection, protect EMDE's from negative spillovers, or preserve regulatory autonomy remain contentious and unresolved. Here, the evidence shows a tension between policymaking practice and trade and investment rules. Further research should continue to explore the environmental and economic impacts of different policies and to assess their relative benefits and drawbacks.

Conclusion

Out of the many potential barriers to effective policy space for GIP that include economic, political, and international determinants, this study focuses on the extent to which trade and investment rules act as a barrier to effective green industrial policymaking. Of the three scenarios laid out in the conceptual framework, scenario 3 seems the most likely: there is heterogeneity in the kinds of policies that advanced economies and emerging market and developing economies are implementing, with the former group 'staying (more) within the rules' than the latter. Consequently, it appears that the rules may constrain one set of countries more than another in

[9]The most frequent type of climate-related ISDS case in recent years has been brought by renewable energy investors in the face of countries reducing incentives for that sector. In these disputes, renewable energy investors in Europe poured in response to new national incentive programs put in place to comply with the EU's renewable energy directive (European Commission, 2009). Unfortunately, due to a confluence of factors (e.g. the Global Financial Crisis and a rapid decrease in the cost of solar market inputs), these new programs became suddenly economically unsustainable, and initially generous investment incentives had to be modified. At the same time as piling-on effects threatened to destabilize these European economies with unsustainable debt, renewable energy producers and investors sued them for rolling back incentives offered initially (Thrasher, 2019). Spain, Italy, Bulgaria, and the Czech Republic have all faced similar suits (e.g. *ACF Renewable Energy v. Bulgaria* (2024), *Canepa v. Spain* (2024), *Encavis v. Italy* (2024), *LSG Building Solutions v. Romania* (2024), *Antaris v. Czech Republic* (2018)).

[10]This effect may also be present in the EU's anti-deforestation law as it impacts both Brazilian export income and the country's commitments under MERCOSUR, a regional trading bloc (Martins, 2024).

GIP. At the same time, the rules do not demand perfect compliance, and some countries may simply ignore them in favour of their policy priorities (and get away with it) because of the disabled WTO Appellate mechanism. Other countries might also be unwilling to challenge those practices because they are engaged in similar ones at home.

In exploring whether there are significant uncontested violations of the rules (scenario 2), the paper finds that, while possible, there are at least a few recent cases in which emerging market and developing economies have been challenged at the WTO for policies that could be included in the GIP dataset – including Turkey's import measures affecting Chinese EVs, Indonesia's export ban on raw materials, and Taiwan's local content requirements in the wind sector.

This tension between policymaking and global rules could be resolved in various ways. One option is that countries change their practices to bring their GIP into conformity with the rules. This approach places a premium on the benefits of legal certainty in international law and may shield more vulnerable countries from the negative economic impacts of the unilateral policies of powerful actors. It has the advantage of not requiring reform of difficult-to-reform global institutions but carries the drawback of forcing countries to slow progress amid a new global consensus in favour of industrial policy.

A second option is to maintain the status quo and engage in rulemaking or enforcement at the levels where it currently takes place – in plurilateral negotiations or sectoral agreements. However, this approach does not attempt large-scale change to the system. Some countries will ignore the rules, and others will not, or will seek to enforce them through existing institutions. This option may allow countries to focus on their own policymaking rather than worry about the status of global rules unless it benefits them. The downside is that it leaves the problematic rules that constrain policymaking to bind the weakest and most vulnerable countries, while the powerful ignore those constraints, further exacerbating existing wealth and development gaps. As such, it risks a de facto degradation from a rules-based trading system to one based on power.

The third option would be to reform the rules to ensure policy space for innovative and iterative GIP, especially for EMDEs. This is consistent both with the way international law naturally evolves and with the urgent demands of the climate crisis. Countries' practices contribute to the development of new international legal norms. As policymaking shifts in favour of GIP, it is natural that the rules should evolve as well. In addition, the need for ambitious climate action points to the importance of special treatment for GIP under trade and investment rules that is facilitative rather than prohibitive. The rules must make space for policies that help countries achieve their individual climate goals, while encouraging cooperation and coordination between countries and accounting for different levels of fiscal space, development capacity, and climate vulnerability. Determining how to change the rules to achieve these ambitious goals simultaneously should be the focus of future research.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The data that support the findings of this study are available from the corresponding author, [PB], upon reasonable request.

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